Venkata Lakshmi Parimala Pasupuleti

AI Application Software Engineer · Python / LangGraph / LLM Integration · Observability & Evaluation for AI Systems · Event-Driven Microservices

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SUMMARY

AI Application Software Engineer with 9 years of production backend engineering and hands-on LLM integration experience. Built AgentProbe -- a production observability and monitoring framework for AI applications: 5-metric eval engine, CI-integrated regression gate, LLM-as-judge scoring, agent tracing, and multi-provider LLM routing across Anthropic, OpenAI, Groq, and Ollama. Built LeaseI -- an 8-agent LangGraph AI application with RAG pipelines, vector database retrieval, tool calling, and HITL guardrails, evaluated across 1,240 test cases. Seven years at Entrata owning scalable distributed systems in Python, Java, and PostgreSQL -- including event-driven async architecture (RabbitMQ), microservices (Docker, K8s), and 200+ SQL query optimizations. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & LLM Integration: LangGraph, LangChain, LLM Integration, AI Agents, RAG Pipelines, Vector Database (ChromaDB), Embedding Models, Semantic Search, Prompt Engineering, Tool Calling, HITL, Guardrails, LLM Evaluation, LLM-as-Judge, Observability for AI, Agent Tracing, Hallucination Detection, Regression Suites, CI Eval Gate, OpenAI API, Anthropic API, Groq, Ollama, MCP

Languages & Frameworks: Python, Java, JavaScript, TypeScript, SQL, PL/SQL, PHP, R, Bash, HTML/CSS, FastAPI, REST APIs, React.js, Angular, Node.js, Spring Boot, .NET Framework

Data & ML: PostgreSQL, MySQL, Oracle, NoSQL, ETL, Data Modeling, Data Pipelines, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, spaCy, BERT, Hugging Face, Tableau, Power BI

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), RabbitMQ (event-driven architecture), Redis, Microservices, Distributed Systems, System Design, Performance Engineering, Production Monitoring, Git, Linux, Agile/Scrum, Libsodium

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Integrated LLM capabilities into a production agentic system using LangGraph -- parse → plan → retrieve → validate → conditional routing; built RAG layer over regulatory corpus answering queries with cited sources; implemented validation guardrails blocking non-compliant outputs before export.

Entrata (PropTech / Real Estate SaaS · 100K+ housing units · 100K+ MAU)
Sep 2018 - Aug 2024

Senior Software Engineer

- Built and owned RabbitMQ-based event-driven async architecture for gross-rent updates across thousands of tenant records -- designed retry logic and idempotency guarantees that eliminated double-processing failures; owned system through production incidents and annual schedule changes.
- Shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints; led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction); owned backend services and application logic end-to-end from design through delivery.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning; owned database schemas and data models for eligibility, financial, and document workflows across multi-tenant production environments.
- Led AWS S3 document migration -- 100% PHI compliance with Libsodium-encrypted pipelines, zero security incidents; built Python/Bash automation pipelines eliminating 80% of manual operational workload.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools reducing debugging time by 40%.

Entrata India (Db Xento Systems)
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee Software Engineer

- Built property management modules in Python, Java, and PostgreSQL; led AWS S3 cloud migration; redesigned document storage database architecture migrating hundreds of thousands of records to a modern schema.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source prediction models (Python, R) -- $R^2 = 0.977$ accuracy; applied Isolation Forest for automated data integrity checks; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Application Observability & Evaluation Framework
2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions)

- Built a production observability and monitoring framework for AI applications: 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking quality degradation on every commit, LLM-as-judge scoring with multi-trial

pass@k harness, agent tracing, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama). 15-endpoint FastAPI backend; evaluated LeaseI across 1,240+ test cases.

LeaseI: Production LLM Application with RAG & Agent Framework

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · Vector Database · OCR · SQLite

- Built an 8-agent LangGraph AI application integrating LLMs, RAG pipelines, vector database retrieval, tool calling, and intelligent automation workflows; HITL guardrails on regulated decisions; 93.6% precision across 1,240 test cases; full observability through AgentProbe eval harness.

AI on FHIR: NLP-Driven Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- Integrated LLM + BERT + embedding models into a production Flask + React application; reduced manual retrieval time ~60%; HIPAA-compliant RBAC; NLP-to-FHIR API query translation in EN/ES.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017